

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 2 – Technical Presentation

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 2: Technical Presentation
Weightage	30% of CIA	Total Marks	100
CO1 Statement	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2,BL3).		
Student Name	D.Mahaboob Basha	Reg. No.	192525369
Section/Batch	Slot C / 2025	Date	06-10-2026

Code of Conduct:

I, D Mahaboob Basha(192525369), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D Mahaboob Basha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



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TECHNICAL PRESENTATION

Stack ADT-Based Backward Navigation in Web Browsers

CSA0314-DATA STRUCTURES

GUIDED BY:
Dr Krishnaraj M
Associate Professor

PRESENTED BY:
D Mahaboob Basha
192525369(AIML)

INTRODUCTION

- Web browsers allow users to visit multiple web pages.
- Backward navigation helps users return to previously visited pages.
- Stack ADT(Abstract Data Type) can be used to explain browser history management
- Google Chrome is a real-world example

Problem Statement

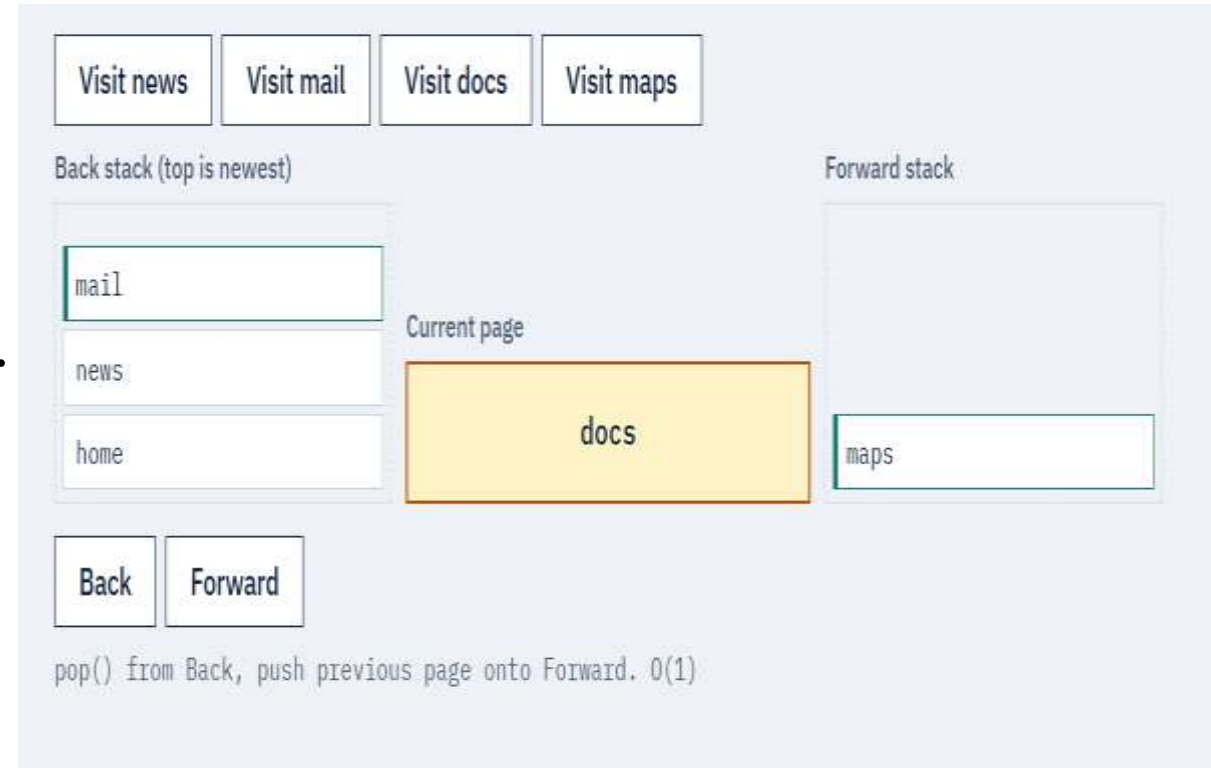
- Users visit multiple web pages in sequence.
- They may need to return to a previously visited page.
- Searching the entire browsing history is unnecessary for the immediate previous page.
- An efficient method is needed to manage navigation history.

Operations in Stack ADT

- Stack is a linear data structure.
- It follows LIFO (Last In, First Out).
- The last inserted element is removed first.

Basic operations:

- Push: Add an element.
- Pop: Remove the top element.
- Peek: View the top element.
- isEmpty: Check whether the stack is empty.



Performance Analysis of Stack

Operation	Time Complexity
Push	$O(1)$
Pop	$O(1)$
Peek	$O(1)$
Display	$O(n)$
Space Complexity	$O(n)$

$O(1)$: Constant time; the operation does not depend on the number of stored pages.

$O(n)$: Linear time; the time or space grows proportionally with the number of pages.

Applications

- Web browsers.
- Undo and redo operations in text editors.
- Mobile application navigation
- Function calls in programming

Conclusion

- Stack ADT is a suitable model for understanding backward navigation in web browsers.
- It follows the LIFO principle, making it easy to retrieve recently visited pages first.
- Stack operations like Push and Pop are efficient, typically taking $O(1)$ time.
- Stacks help organize data and are widely used in browsers, programming, and other applications.
- Understanding Stack ADT shows how simple data structures improve performance, organization, and user experience.

THANK YOU